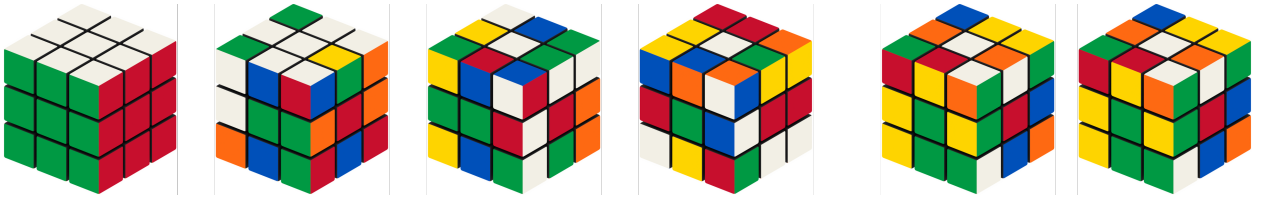
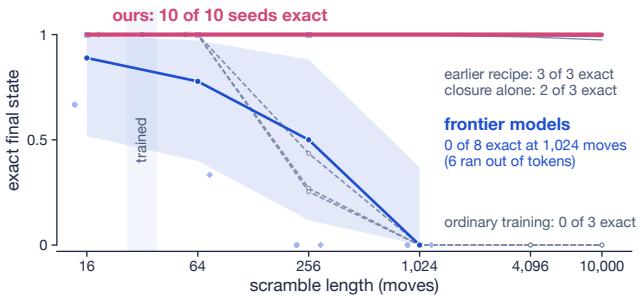
a **Trained on scrambles of 32 moves, the tracker follows one scramble for 4,096 moves and reads out the true cube**



b **Exact to 10,000 moves; the frontier models fail by 1,024**



c **Frontier tokens per problem; ours generates none**

